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Increased serum amyloid A and urinary kidney injury marker - 1 concentrations in cats with pyelonephritis: a diagnostic accuracy study

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Abstract

Objectives

To investigate the discriminatory potential of serum amyloid A, neutrophil:lymphocyte ratio, monocyte:lymphocyte ratio and assess renal tubular injury in cats with pyelonephritis.

Materials and Methods

Diagnostic accuracy study using samples from prospectively enrolled cats and stored material. Cats were classified according to diagnosis as pyelonephritis, ureteral obstruction, chronic kidney disease, subclinical bacteriuria, cystitis, or healthy. Serum amyloid A, neutrophil:lymphocyte ratio, monocyte:lymphocyte ratio and urinary kidney injury marker-1 were compared across groups. Receiver operating characteristic curve analysis assessed the performance of inflammatory

biomarkers. Serum amyloid A and urinary kidney injury marker-1 were measured consecutively in a subset of cats with pyelonephritis.

Results

Samples from 71 cats (pyelonephritis $n = 22$, other urological conditions $n = 35$, healthy $n = 14$) were included. Median serum amyloid A concentration was significantly higher in cats with pyelonephritis, overlapping minimally with other groups. At a threshold of 49.1 mg/L, serum amyloid A showed a positive likelihood ratio of 32, sensitivity of 0.95 (95% CI: 0.74 to 0.99) and specificity of 0.97 (95% CI: 0.85 to 0.99), outperforming neutrophil:lymphocyte ratio. Serum amyloid A normalised in 12/14 cats post-treatment. Urinary kidney injury marker-1 was elevated in more than 50% of cats with pyelonephritis and ureteral obstruction, significantly higher than in healthy cats and those with subclinical bacteriuria or cystitis. Variation in urinary kidney injury marker-1 was observed during follow-up.

Conclusions

At a threshold of 49 mg/L, serum amyloid A is a highly accurate biomarker for differentiating pyelonephritis from other urological conditions. Elevated urinary kidney injury marker-1 suggests frequent proximal tubular injury.

Clinical Significance

Pyelonephritis is challenging to diagnose. Serum amyloid A can improve diagnostic precision and support timely treatment decisions. Urinary kidney injury marker-1 helps identify tubular injury; however, the clinical implications remain to be investigated.

INTRODUCTION

Pyelonephritis (PN) is an infection of the renal pelvis and parenchyma, most commonly resulting from ascending bacteria originating in the lower urinary tract. Like other urinary tract infections in cats, PN is mostly seen in elderly female cats, and underlying kidney disease appears to be common (Whitehouse et al., [2024](#)). Despite its potential severity, the clinical presentation of PN can be non-specific, and the diagnosis is rarely straightforward (Weese et al., [2019](#)). To obtain a diagnosis with certainty, a positive culture on urine obtained by pyelocentesis is required, a technically challenging procedure beyond routine reach. Thus, in most clinical settings, the

diagnosis relies on a positive bladder urine culture coupled with clinical, laboratory and/or ultrasonographic signs suggestive of upper urinary tract infection.

Unfortunately, PN shares numerous characteristics with other urological disorders, especially chronic kidney disease (CKD) and ureteral obstruction (UO), resulting in considerable diagnostic overlap. Additionally, cases involving coexisting urological conditions, such as subclinical bacteriuria (SBU) or cystitis in cats with CKD, are not uncommon and are particularly complex to differentiate from PN.

Serum amyloid A (SAA) is a major acute-phase protein which increases rapidly in response to an inflammatory stimulus. Serum amyloid A is produced in the liver, and increases are detectable in blood within 8 hours (Rossi, [2023](#)). High magnitude increases of SAA have been demonstrated in cats with inflammatory conditions such as upper respiratory disease, pneumonia, pyometra, feline infectious peritonitis and sepsis (Troia et al., [2017](#); Yuki et al., [2020](#)). Recently, significant increases in SAA concentrations were also demonstrated in 17 cats with pyelonephritis when compared to cats with other urological conditions (Kurtz et al., [2024](#)), suggesting that SAA might be a promising discriminatory biomarker in this population of cats.

Neutrophilia, monocytosis and lymphopaenia are common haematological findings in cases of systemic inflammation. Neutrophil:lymphocyte ratio (NLR) and monocyte:lymphocyte ratio (MLR) can be considered as indicators of the immune-inflammatory response. They are easily calculated ratios, combining cellular parts of the innate and adaptive immune system, with the components available from a routine complete blood count (CBC) as performed by most automated laboratory haematological analysers. Alterations of NLR and MLR have been demonstrated in cats with various inflammatory and infectious conditions (Donato et al., [2023](#); Gori et al., [2021](#)), and leukocyte ratios could potentially serve as discriminatory markers in cats with PN.

Kidney injury marker-1 (KIM-1) is a scavenger receptor of the proximal tubular epithelial cells and a highly specific marker of ischaemic or toxic acute kidney injury in people, cats and rodents. KIM-1 is well conserved across species, and the KIM-1 molecule has been identified in urine and renal tissue of cats with confirmed and suspected acute kidney injury (Bland et al., [2014](#); Bland et al., [2017](#)). The extent of tubular damage in cats with PN has not been investigated.

The primary aim of this study was to investigate and assess the discriminatory potential of selected non-invasive inflammatory biomarkers, SAA, NLR and MLR. We hypothesised that cats with PN would have systemic inflammation and thus increased concentration of SAA as well as

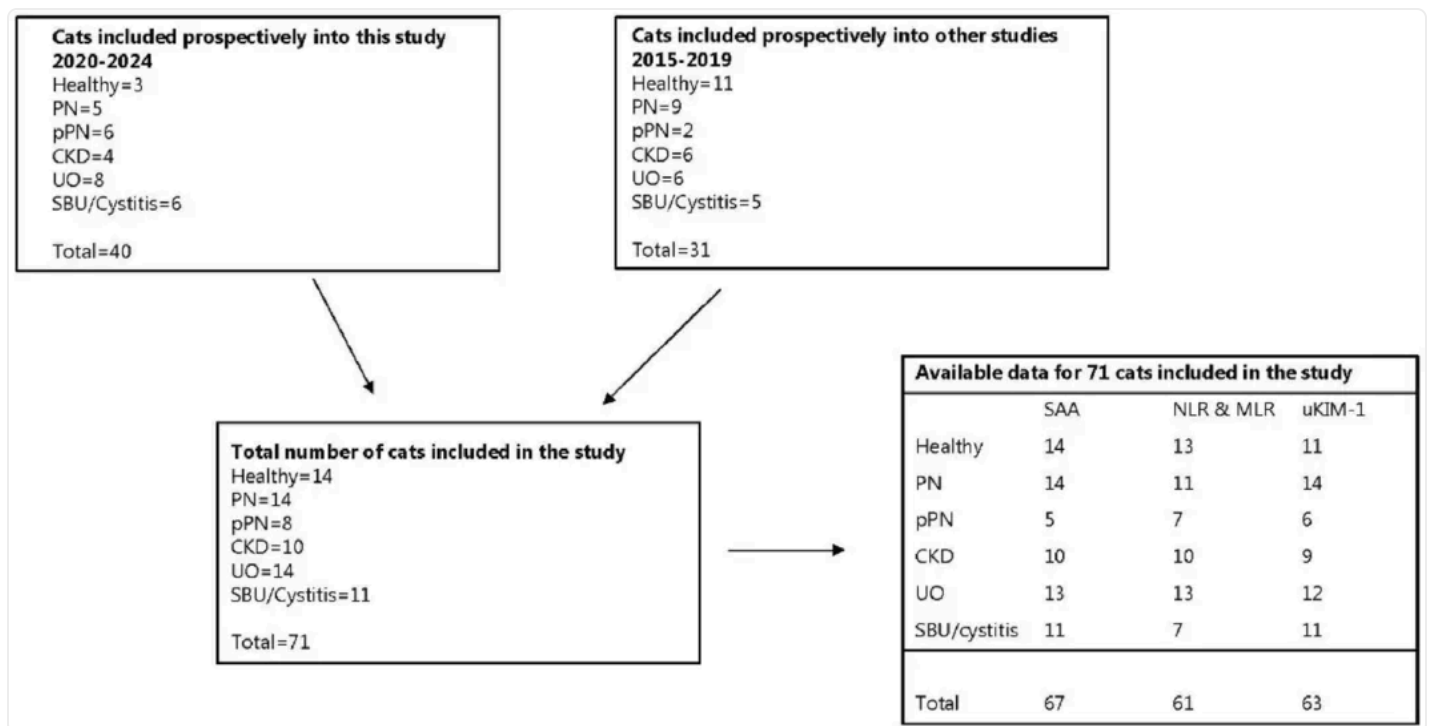
higher NLR and MLR than cats with other urological conditions, such as UO, CKD, cystitis or SBU or a combination of these. We also hypothesised that SAA would decrease in response to therapy in cats with PN. The secondary aim was to investigate renal tubular injury in cats with PN and to assess whether urinary KIM-1 (uKIM-1) can serve as a biomarker to identify cats with PN, discriminating these from cats with other urological conditions. We hypothesised that cats with PN and cats with UO are at risk for proximal tubular injury and thus would have higher uKIM-1 concentrations compared to healthy cats, cats with bacterial infection (cystitis) or colonisation (SBU) of the lower urinary tract, and cats with CKD.

MATERIALS AND METHODS

Study design and enrolment of cats

The study was designed as a diagnostic accuracy study, with the reporting following the STARD criteria (Table [S1](#)). Cats were either prospectively enrolled into this study from 2020 to 2024 or enrolled into previous studies at the University Hospital for Companion Animals (UHCA) at the University of Copenhagen, Denmark from 2015 to 2019 (Godia et al., [2022](#); Jessen et al., [2020](#); Langhorn et al., [2019](#)) with stored urine and blood samples and medical record data included in the present study (Fig [1](#)). Cats with urological conditions were a convenience sample consisting of cats presenting to the UHCA or Blue Star Animal Hospital in Gothenburg (BSAH), Sweden, and fulfilling the criteria described below, whereas healthy control cats were recruited for the purpose of research. Recruitment took place at the UHCA, except for three cats that were recruited from the BSAH. Ethical permission was obtained from the Ethical Committee at the UHCA. Informed consent was obtained from owners of all cats participating in this study.

FIG 1.



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Number of cats included and data available for analysis of serum amyloid A, neutrophil:lymphocyte ratio, monocyte:lymphocyte ratio and urinary kidney injury marker-1. CKD Chronic kidney disease; PN Pyelonephritis; pPN Presumptive pyelonephritis; SBU Subclinical bacteriuria; UO Ureteral obstruction.

Classification of cats

Cats were classified as PN (confirmed PN) based on a positive urine culture obtained by pyelocentesis. Cats were classified as pPN (presumptive PN) based on clinical AND/OR clinicopathological findings suggestive of PN in combination with a positive urine culture obtained by cystocentesis. Clinical and clinicopathological findings suggestive of PN were defined as (i) signs of systemic inflammation such as pyrexia AND/OR unspecific signs of systemic illness such as lethargy/inappetence/vomiting AND/OR haematological signs of systemic inflammation such as neutrophilia, neutropenia, presence of band neutrophils AND (ii) signs of renal involvement (renal pain AND/OR azotaemia). In previous studies aimed at evaluating the microRNAome in cats

with pyelonephritis (Godia et al., [2022](#); Jessen et al., [2020](#)), increased acute phase proteins were part of the inclusion criteria for cats with pPN. Cats with pPN included before 2020 were, therefore, excluded from the evaluation of SAA as a predictive marker in this study.

Cats were classified as having other urological conditions if diagnosed with either CKD, UO, cystitis, SBU, or a combination of these. A diagnosis of CKD required stable azotaemic renal disease (defined as serum creatinine above 174 $\mu\text{mol/L}$) with a urine specific gravity (USG) below 1.035, or degenerative changes on ultrasonography, and a negative cystocentesis urine bacterial culture. For cats that were retrospectively included from previous prospective urological research projects, a diagnosis of CKD made by the researchers of the previous study was considered acceptable if the medical record was not available for evaluation. UO (complete or partial) was diagnosed if cats had uni- or bilateral ureteral obstructive disease on ultrasonographic evaluation with a concurrently negative pyelocentesis urine bacterial culture. Cats were diagnosed with cystitis when presented with clinical signs of lower urinary tract infection (pollakiuria, dysuria, stranguria, periuria and/or haematuria) in the presence of a positive urine culture obtained by cystocentesis, whereas SBU was recorded if a positive cystocentesis urine culture was obtained in the absence of these clinical signs. Finally, clinically healthy cats ≥ 1 year of age were classified as healthy based on an unremarkable clinical examination and clinicopathological findings (complete blood count, biochemistry and urinalysis). For all cats, it was an exclusion criterion if the cats had been receiving antibiotics ≤ 72 hours before evaluation for study inclusion.

Diagnostic work - up

Diagnostic work-up of animals with PN, pPN and cats with other urological conditions was performed according to the discretion of the attending clinician, except for post-treatment SAA and uKIM-1 analysis, which was performed for the purpose of the study.

For each cat, a CBC, serum biochemistry, urinalysis and urine culture were performed at the Veterinary Diagnostic laboratory, University of Copenhagen, Denmark. Quantitative aerobic urine culture was performed at SUND Vet Diagnostik, University of Copenhagen, Denmark and at the BSAH diagnostic laboratory, Sweden. Bacterial species identification was performed by matrix-assisted laser desorption/ionisation-time of flight (MALDI-TOF) mass spectrometry (Vitek MS RUO; BioMérieux). For cats with PN and UO, abdominal ultrasonography was also performed, including the collection of urine by pyelocentesis in case of pyelectasia (>2 mm dilation) for aerobic culture.

NLR and MLR

A CBC was analysed at the time of inclusion. Depending on location and time of admission, analysis was performed with either the Advia 2120i (Siemens Healthineers, Erlangen, Germany) or the Procyte R (IDEXX Laboratories, Inc., Westbrook, Maine). Neutrophilia was defined as $>12.6 \times 10^9/L$ using either instrument, and left shift was defined as an increased band neutrophil count $>0.3 \times 10^9/L$ (confirmed microscopically). Neutrophil:lymphocyte and monocyte:lymphocyte ratios were calculated for each cat.

Serum amyloid A

Serum amyloid A was measured at the time of inclusion as part of the biochemical profile. The analysis was run within 48 hours of sample collection, except for the three cats from BSAH, for which SAA was measured in serum stored at -80°C for 9 years after collection. In cats with PN, SAA was measured again within 1 week of treatment cessation. During treatment, SAA measurement was not part of the protocol but could be performed at the discretion of the attending clinician. The concentration of SAA was analysed at the Veterinary Diagnostic Laboratory at the UCPH (LZ-SAA (Eiken chemicals A/S, Tokyo, Japan) on Advia 1800, SIEMENS, before June 26, 2022, then VET-SAA (Eiken chemicals A/S, Tokyo, Japan) on Atellica analyser, SIEMENS). Increased SAA was defined as a concentration $> 5 \text{ mg/L}$ using either instrument. The microbiologists performing cultures were blinded to the result of the SAA analysis, whereas attending clinicians and investigators were not blinded.

Urinary KIM - 1

Excess urine was aliquoted in 1 mL plastic vials and stored at -80°C . Samples were batched and shipped on ice with courier to the University of Guelph, Canada. Sample identity was disguised before shipping. For analysis, urine samples were thawed and briefly centrifuged. A $20 \mu\text{L}$ aliquot of urine was mixed with $400 \mu\text{L}$ of buffer, and $150 \mu\text{L}$ of this mixture was applied to a feline-specific lateral flow assay. After 15 minutes of incubation at room temperature, the KIM-1 of the test sample and of the positive control impregnated into the membrane was measured with a handheld radiofrequency identification reader (Bland et al., [2019](#)). The ratio of the test to control sample optical density was calculated and normalised to a USG of 1.035.

Statistical analysis

Statistical analyses were performed using MedCalc® Statistical Software version 22.016 (MedCalc Software Ltd, Ostend, Belgium; <https://www.medcalc.org>; 2023) and GraphPad Prism 10.0.3.

Statistical significance was set at $P < .05$. Tests for normality and log normality were performed using D'Agostino and Pearson tests.

Continuous variables (SAA, NLR, MLR) are presented as medians with 25th and 75th percentiles. A Kruskal-Wallis test was applied to detect statistically significant differences between groups of cats, followed by Dunn's *post hoc* analysis. In one cat with pPN and in three cats with other urological disorders, SAA concentrations were below the lower reporting limit (<5 mg/L) of the VET-SAA assay, and the SAA concentrations were assigned a value of 0.

Receiver operating characteristic analysis was employed to calculate the area under the curve (AUC) of SAA, NLR and MLR to differentiate between cats with PNs (confirmed and presumptive) and cats with other urological conditions. The 95% confidence interval (95% CI) was calculated. To obtain estimates of the optimal cut-off value for SAA, LNR and MLR, the cut-off value associated with the maximum value of the Youden index ($\text{sensitivity}_c + \text{specificity}_c - 1$; with c representing all possible cut-off values) was assigned as the optimal cut-off value. Because the optimal cut-off value associated with the maximum Youden index relies on a disease prevalence of 50%, giving equal weight to sensitivity and specificity and ignoring costs of various decisions, the optimal cut-off value was also estimated using the cost approach in MedCalc. Briefly, sensitivity (SE), specificity (SP) and positive likelihood ratio (LR+) were calculated for disease prevalences of 10%, 33% and 50% and assuming costs of false positive and false negative of 1 compared to costs of true positive and true negative of 0.

RESULTS

A total of 71 cats were included in the study (Fig [1](#)). Demographic data, key clinical features and routine laboratory data indicative of systemic inflammation and renal involvement are listed in Tables [1](#) and [2](#), respectively. Comorbidities and underlying abnormalities in cats participating in the study are listed in Table [S2](#). Stratified data for cats with other urological conditions are shown in Table [S3](#).

Table 1.

Demographic data for the 71 included cats

	PN	pPN	Other	Healthy cats
<i>n</i>	14	8	35	14
Breeds	Abyssinian (1)	Birman (3)	Abyssinian	DSH (10)
	Bengal (1)	Burmilla (1)	(1)	Norwegian forest cat (2)
	Birman (1)	DSH (4)	Birman (2)	Ragdoll (2)
	DSH (5)		British shorthair (1)	
	Maine Coon (2)		DLH (2)	
	Oriental (2)		DSH (20)	
	Ragdoll (1)		Maine Coon mix (2)	
	Siberian forest cat (1)		Maine Coon (2)	
			Norwegian forest cat (1)	
			Ocicat (1)	
			Oriental (1)	
			Ragdoll (1)	
			Unknown (1)	

	PN	pPN	Other	Healthy cats
Age (years)	8 (2 to 14)	8.5 (4 to 17)	8 (1 to 17)	6 (2 to 16)
Sex	FE (2)	FE (2)	FE (4)	FE (2)
	FN (12)	FN (4)	FN (14)	FN (7)
	ME (0)	ME (0)	ME (0)	ME (1)
	MN (0)	MN (2)	MN (17)	MN (4)

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Ages are presented as median (range)

FE Female entire, FN Female neutered, ME Male entire, MN Male neutered, PN

Pyelonephritis, pPN Presumptive pyelonephritis

Table 2.

Key features of 14 cats with confirmed PN, 8 cats with pPN, 35 cats with other urological diseases and 14 healthy cats

	PN	pPN	Other	Healthy cats
Abdominal pain	7/13, 53.8%	4/8, 50%	5/34, 14.7%	0/14, 0%
Rectal temperature	39.3 (37.4 to 40.4)	39.25 (37.4 to 40.5)	38.1 (37.1 to 39.5)	38.5 (37.6 to 39.2)
Pyrexia (<i>n/n</i> , %)	8/13, 61.5%	3/8, 37.5%	2/19, 10.5%	0/5, 0%
Creatinine ($\mu\text{mol/L}$) (range)	204 (102 to 735)	276.5 (125 to 419)	197 (62 to 1470)	120 (86 to 133)
Creatinine > 140 ($\mu\text{mol/L}$) (<i>n/n</i> , %)	12/13, 92.3%	7/8, 87.5%	25/35, 71.4%	0/14, 0%
Azotaemia (<i>n/n</i> , %)	8/13, 62%	6/8, 75%	20/35, 57%	0/14, 0%
USG	1024.5 (1006 to 1051)	1018.5 (1013 to 1050)	1020.5 (1010 to 1052)	1050 (1042 to 1050)
Neutrophils	14.9 (11.8 to 66.27)	18.22 (4.99 to 33.66)	5.67 (1.56 to 11.57)	4.19 (1.63 to 6.76)
Neutrophilia (<i>n/n</i> , %)	10/11, 91%	5/8, 62.5%	0/32, 0%	0/15, 0%
Left shift (<i>n/n</i> , %)	6/11, 54%	4/7, 57.1%	0/29, 0%	0/15, 0%
Uropathogens	<i>E. coli</i> (11) <i>Enterococcus faecalis</i> (1) (this cat also had <i>E. coli</i> in the bladder)	- <i>E. coli</i> (5)	<i>E. coli</i> (5) <i>Staphylococcus felis</i> (1)	

	PN	pPN	Other	Healthy cats
	<i>Staphylococcus felis</i> (1) <i>Staphylococcus</i> spp. (1)	Haemolytic and non-haemolytic <i>E. coli</i> (1) <i>Enterococcus faecalis</i> (1) <i>Enterococcus</i> spp. (1)	<i>Staphylococcus aureus</i> subspecies <i>aureus</i> (1) <i>Staphylococcus</i> spp. (1) <i>Streptococcus</i> spp. (1) <i>Micrococcus luteus</i> (1) <i>Bacillus</i> (1)	

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Azotaemia is defined as creatinine >17 µmol/L. Numerical variables are presented as median values (range). Pyrexia is defined as $T \geq 39.2^{\circ}\text{C}$. USG > 1050 was recorded as 1050 and USG < 1020 was recorded as 1019 in the calculation of the median USG Urine specific gravity

SAA, NLR and MLR

The median SAA concentration was significantly higher in cats with PN (180.3 mg/L) and pPN (607.4 mg/L) when compared to healthy cats (0.0 mg/L) and cats with other urological conditions (0.4 mg/L) (Table 3). There was minimal overlap of SAA values between groups of cats (Fig 2A).

Table 3.

Serum amyloid A, neutrophil:lymphocyte ratio and monocyte:lymphocyte ratio measurements in healthy cats, cats with other urological conditions, cats with pyelonephritis and cats with presumptive pyelonephritis

	PN	pPN	Other	Healthy cats
SAA (mg/L)	<i>N</i> = 14 Median: 180.3 ₋ [*] (167.2 to 205.8)	<i>N</i> = 5 Median: 607.4 ₋ [*] (134.9 to 690.5)	<i>N</i> = 34 Median: 0.4 (0.0 to 3.1)	<i>N</i> = 14 Median: 0.0 (0.0 to 0.2)
NLR	<i>N</i> = 11 Median: 15.20 ₋ [*] (6.36 to 22.75)	<i>N</i> = 7 Median: 11.85 (3.63 to 25.75)	<i>N</i> = 30 Median: 2.33 (1.85 to 5.72)	<i>N</i> = 13 Median: 1.66 (1.52 to 2.54)
MLR	<i>N</i> = 11 Median: 0.31 ₋ [*] (0.12 to 0.65)	<i>N</i> = 7 Median: 0.16 ₋ [*] (0.15 to 0.63)	<i>N</i> = 30 Median: 0.08 (0.06 to 0.23)	<i>N</i> = 13 Median: 0.07 (0.04 to 0.10)

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SAA, NLR and MLR are presented as median (25th and 75th percentile). Median values between groups were statistically assessed by the Kruskal–Wallis test with Dunn's *post hoc* test

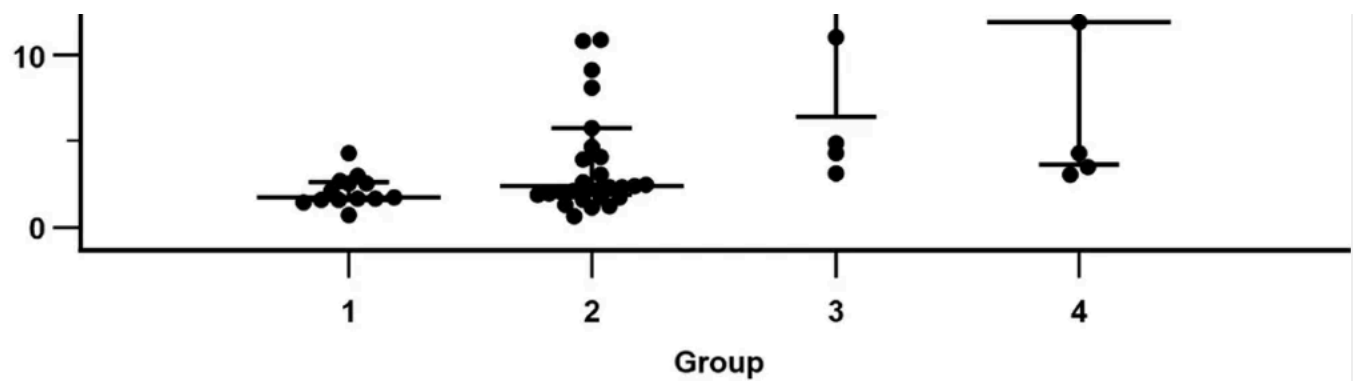
MLR Monocyte:lymphocyte ratio, *N* Number of cats, NLR Neutrophil:lymphocyte ratio, PN Pyelonephritis (confirmed), pPN Presumptive pyelonephritis, SAA Serum amyloid A

*Median concentrations of SAA in cats with confirmed pyelonephritis were significantly higher than those of healthy cats and cats with other urological conditions ($P < .001$).

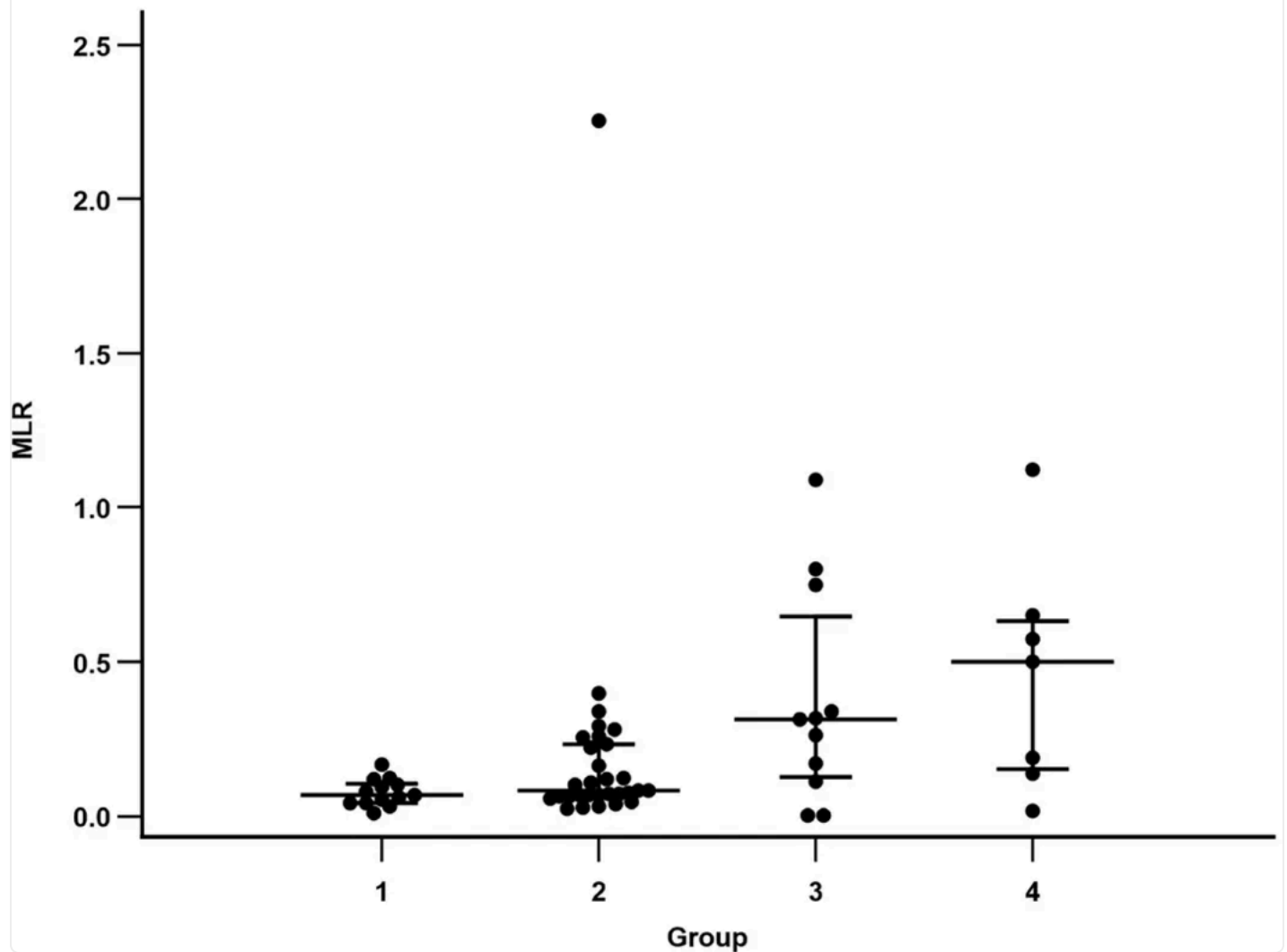
Median concentrations of SAA in cats with presumptive pyelonephritis were significantly

higher than those of healthy cats ($P = .004$) and cats with other urological conditions ($P = .021$). For NLR, cats with confirmed pyelonephritis had significantly higher ratios than healthy cats ($P = .006$). For MLR, cats with confirmed pyelonephritis and cats with presumptive pyelonephritis had significantly higher ratios than healthy cats ($P = .039$)

FIG 2.



C



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(A) Serum amyloid A (SAA) in 14 clinically healthy cats (Group 1), 34 cats with other urological conditions (Group 2), 14 cats with confirmed pyelonephritis (Group 3) and 5 cats with presumptive pyelonephritis (Group 4). Horizontal lines represent medians and 25th to 75th percentiles. Median concentrations of SAA in cats with confirmed pyelonephritis (Group 3) were significantly higher than those in healthy cats (Group 1) and cats with other

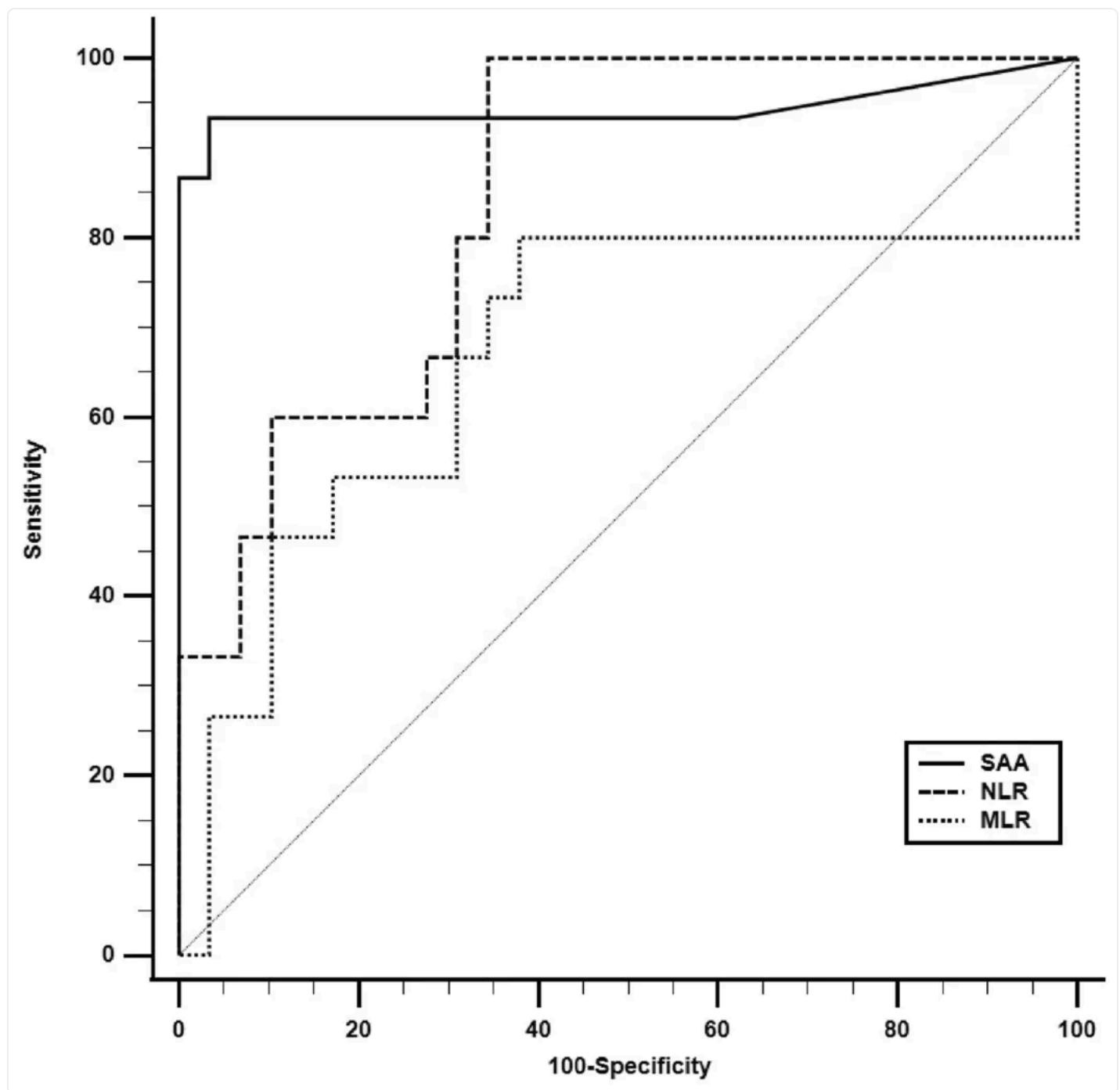
urological conditions (Group 2) ($P < .001$). Median concentrations of SAA in cats with presumptive pyelonephritis (Group 4) were significantly higher than those in healthy cats (Group 1) ($P = .004$) and cats with other urological conditions ($P < .021$). (B) Neutrophil:lymphocyte ratio (NLR) in 13 clinically healthy cats (Group 1), 30 cats with other urological conditions (Group 2), 11 cats with confirmed pyelonephritis (Group 3) and seven cats with presumptive pyelonephritis (Group 4). Horizontal lines represent medians and 25th to 75th percentiles. Cats with confirmed pyelonephritis (Group 3) had significantly higher NLR ratios than healthy cats (Group 1) ($P = .006$). (C) Monocyte:lymphocyte ratio (MLR) in 13 clinically healthy cats (Group 1), 30 cats with other urological conditions (Group 2), 11 cats with confirmed pyelonephritis (Group 3) and seven cats with presumptive pyelonephritis (Group 4). Horizontal lines represent medians and 25th to 75th percentiles. Cats with confirmed pyelonephritis (Group 3) and cats with presumptive pyelonephritis (Group 4) had significantly higher MLR ratios than healthy cats (Group 1) ($P = .039$).

The median NLR was significantly higher in cats with confirmed PN (15.2) when compared to healthy cats (1.66), but not compared to cats with other urological conditions (2.33) (Table [3](#)). The median MLR was significantly higher in cats with PN (0.31) and pPN (0.16) when compared to healthy cats (0.07), but not compared to cats with other urological conditions (0.08) (Table [3](#)). There was substantial overlap of NLR and MLR between groups of cats (Fig [2B,C](#)).

Stratified data for cats with other urological conditions are shown in Table [S4](#).

Receiver operating characteristic curve analysis (Fig [3](#)) revealed that SAA and NLR discriminated between cats with PN (confirmed and presumptive) and cats with other urological conditions, with SAA having superior accuracy to NLR. Area under curves, optimal cut-off, sensitivity, specificity and positive likelihood ratios for SAA, NLR and MLR are listed in Table [4](#), and disease prevalence-directed cut-offs are shown in Table [5](#).

FIG 3.



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Receiver operating characteristic (ROC) curve for serum amyloid A (SAA), neutrophil:lymphocyte ratio (NLR) and monocyte:lymphocyte ratio (MLR) in cats with pyelonephritis and presumptive pyelonephritis *versus* cats with other urological conditions.

Table 4.

Comparison of the area under receiver operating characteristic curves for serum amyloid A, neutrophil:lymphocyte ratio and monocyte:lymphocyte ratio in distinguishing between cats with confirmed and presumptive pyelonephritis and cats with other urological conditions, likely to be clinically relevant differential diagnoses to pyelonephritis

	AUC (95% CI)	Cut-off	Sens (95% CI)	Spec (95% CI)	LR+
SAA	0.957 (0.862 to 0.994)	49.1	0.95 (0.74 to 0.99)	0.97 (0.85 to 0.99)	32.2
<i>N</i> = 19 (PN and pPN) versus 32 Other					
NLR	0.867 (0.737 to 0.947)	3.01	1.00 (0.82 to 1.00)	0.63 (0.44 to 0.80)	2.7
<i>N</i> = 18 (PN and pPN) versus 30 Other					
MLR	0.702 (0.572 to 0.840)	0.29	0.56 (0.31 to 0.79)	0.90 (0.74 to 0.98)	5.6
<i>N</i> = 18 (PN and pPN) versus 30 Other					

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The group of cats with pyelonephritis consisted of cats with a confirmed diagnosis of pyelonephritis (*n* = 11) and cats with presumptive pyelonephritis (*n* = 8)

AUC Area under the curve, LR+ Positive likelihood ratio, MLR Monocyte:lymphocyte ratio, NLR Neutrophil:lymphocyte ratio, pPN Presumptive pyelonephritis, PN Confirmed pyelonephritis, SAA Serum amyloid A, Sens Sensitivity, Spec Specificity

Table 5.

Optimal cut-off value for serum amyloid A, neutrophil:lymphocyte ratio and monocyte:lymphocyte ratio in distinguishing between cats with pyelonephritis and cats with other urological conditions likely to be clinically relevant differential diagnoses to pyelonephritis

Prevalence (%)	Optimal cut-off value based on disease prevalence and costs [*]	Optimal cut-off value based on Youden index and assuming that the ratio of cats with pyelonephritis to cats with relevant differential diagnoses reflects the disease prevalence (29%)
SAA		
10	Cut-off value: 143.9 mg/L SE: 0.90 [0.67 to 0.99] SP: 1.00 [0.90 to 1.00]	Cut-off value: 49.1 mg/L SE: 0.95 [0.74 to 0.99] SP: 0.97 [0.85 to 0.99]
33	Cut-off value: 143.9 mg/L SE: 0.90 [0.67 to 0.99] SP: 1.00 [0.90 to 1.00]	
50	Cut-off value: 49.1 mg/L SE: 0.95 [0.74 to 0.99] SP: 0.97 [0.85 to 0.99]	
NLR		
10	Cut-off value: 18.49 SE: 0.39 [0.17 to 0.64]	Cut-off value: 3.01 SE: 1.00 [0.82 to 1.00]

Prevalence (%)	Optimal cut-off value based on disease prevalence and costs [*]	Optimal cut-off value based on Youden index and assuming that the ratio of cats with pyelonephritis to cats with relevant differential diagnoses reflects the disease prevalence (29%)
MLR	33 SP: 1.00 [0.88 to 1.00] Cut-off value: 10.83 SE: 0.67 [0.41 to 0.87]	SP: 0.63 [0.44 to 0.80]
	50 SP: 0.90 [0.74 to 0.98] Cut-off value: 3.01 SE: 1.00 [0.82 to 1.00]	
	SP: 0.63 [0.44 to 0.80]	
	10 Cut-off value: 0.39 SE: 0.39 [0.17 to 0.64]	
	33 SP: 0.97 [0.83 to 0.99] Cut-off value: 0.29 SE: 0.56 [0.31 to 0.79]	
	50 SP: 0.90 [0.74 to 0.98] Cut-off value: 0.29 SE: 0.56 [0.31 to 0.79]	
	SP: 0.90 [0.74 to 0.98]	

Values in brackets are the associated 95% confidence intervals

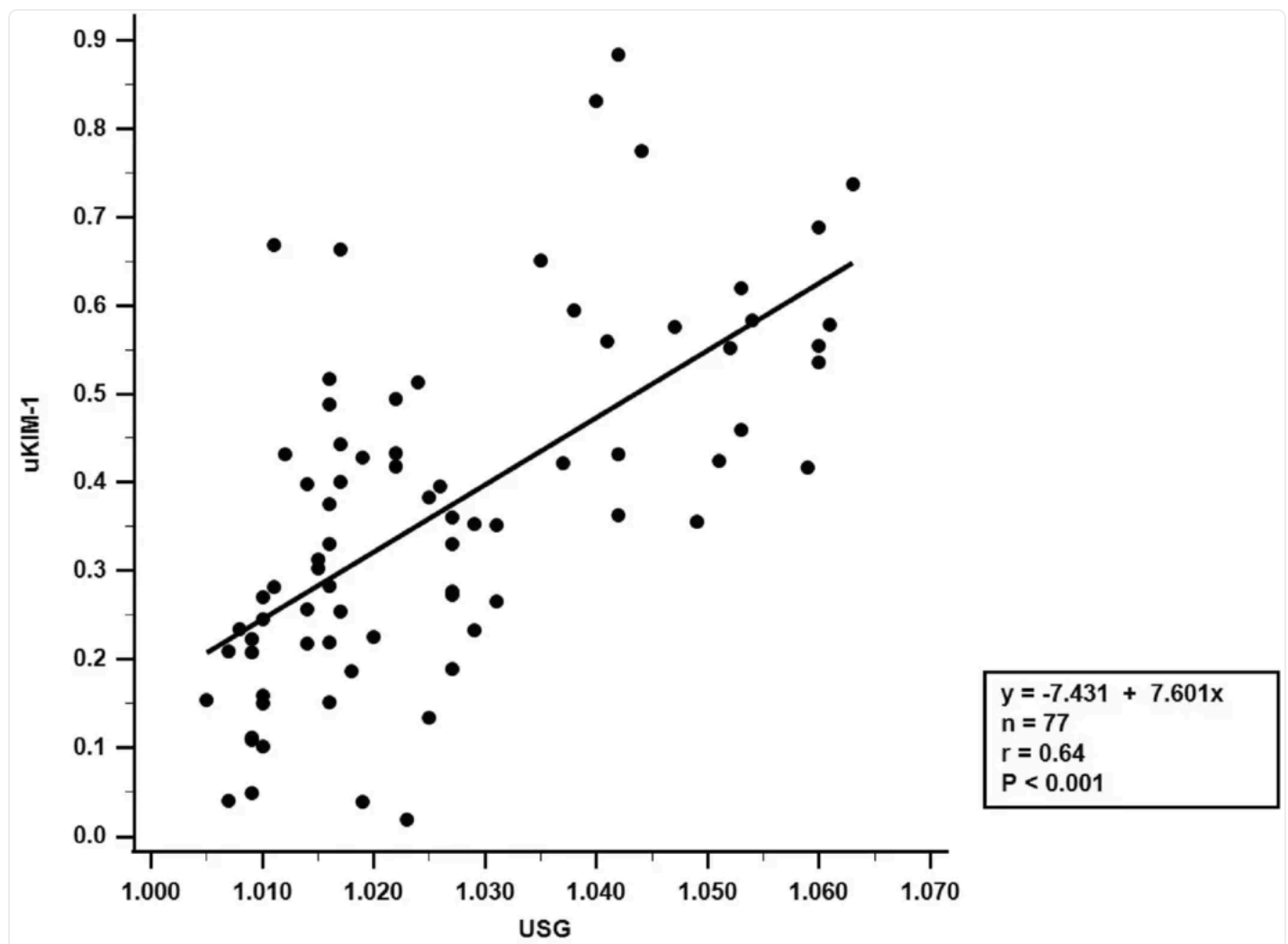
MLR monocyte:lymphocyte ratio, NLR neutrophil:lymphocyte ratio, SAA Serum amyloid A, SE Sensitivity, SP Specificity

*Based on the disease prevalence and estimated cost of false positive: 1; cost of false negative: 1; cost of true positive: 0; and cost of true negative: 0

Urinary KIM - 1

KIM-1 analyses were performed on stored urine samples from 63 cats. The correlation between USG and uKIM-1 is shown in Fig 4. When compared to the 90th percentile of healthy cats, mild to marked uKIM-1 increases were detected in 12/20 and 7/12 cats with pyelonephritis and UO, respectively. Urinary KIM-1 was significantly higher in cats with PN and pPN and in cats with UO compared to healthy cats and cats with SBU/cystitis. There was no statistically significant difference between uKIM-1 in cats with CKD and any other group (Table 6, Fig 5).

FIG 4.



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Correlation between urinary specific gravity and urinary kidney injury marker-1 in 63 cats.

Table 6.

Urinary kidney injury marker-1 in healthy cats, cats with pyelonephritis (confirmed and presumptive) and cats with chronic kidney disease, ureteral obstruction and cystitis/subclinical bacteriuria

	Healthy cats (<i>n</i> = 11)	PN/pPN (<i>n</i> = 20)	CKD (<i>n</i> = 9)	UO (<i>n</i> = 12)	Cystitis/SBU (<i>n</i> = 11)
uKIM-1 (T/C)	0.36 [0.31 to 0.42]	0.73 [*] ₋ [0.49 to 0.86]	0.62 [0.54 to 0.67]	0.76 [*] ₋ [0.49 to 0.93]	0.33 [0.29 to 0.42]

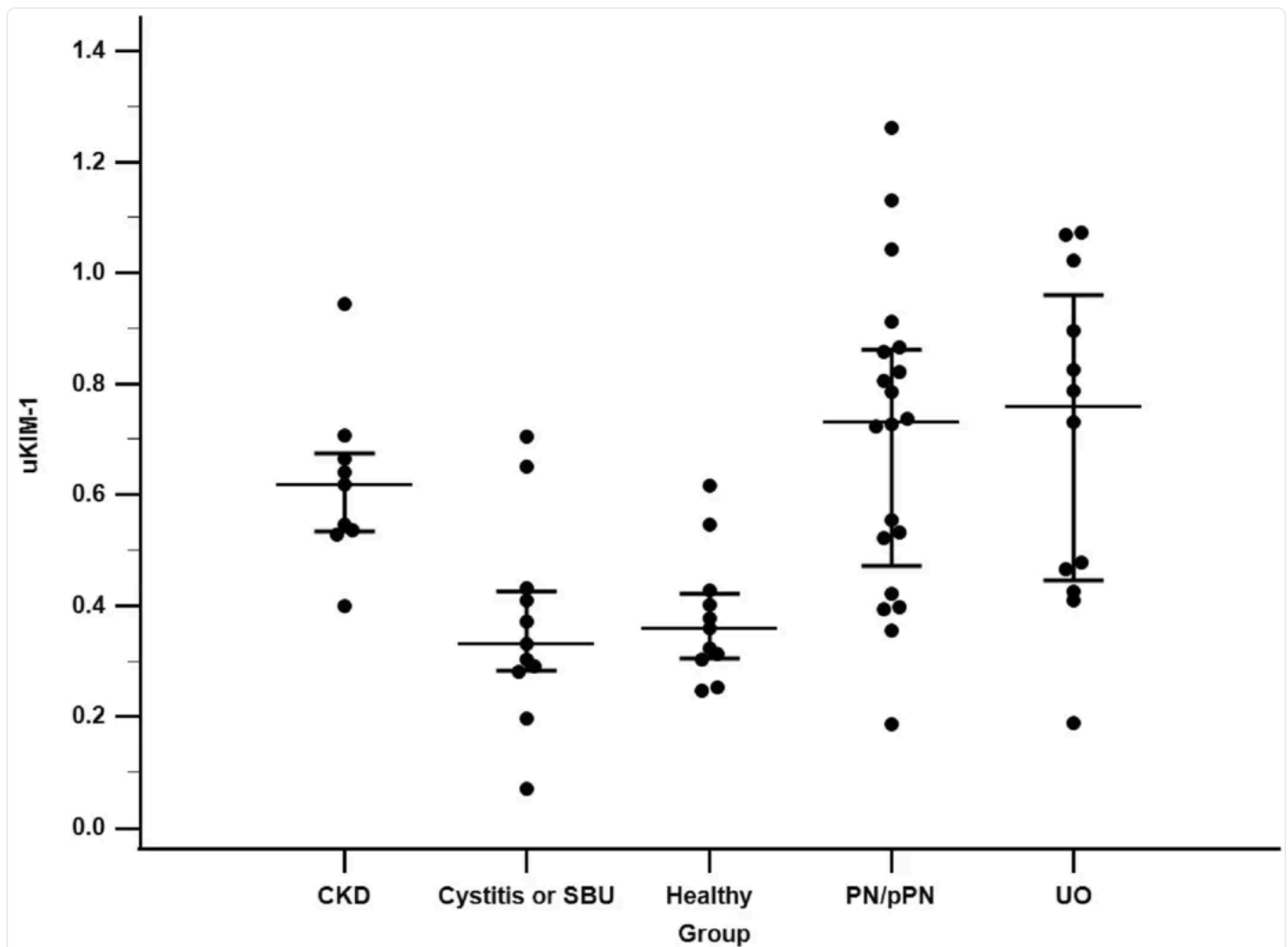
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Numbers are presented as median (25th to 75th percentile). Urinary KIM-1 values are normalised to a urinary specific gravity of 1035 and expressed as test/control ratios; therefore, being unitless

CKD Chronic kidney disease, PN Confirmed pyelonephritis, pPN Presumptive pyelonephritis, SBU Subclinical bacteriuria, T/C Test/control ratio, UO Ureteral obstruction

*Median values between groups were assessed by the Kruskal-Wallis test with Dunn's *post hoc* test indicating statistically significant differences between PN/pPN and healthy cats ($P = .009$), PN/pPN and cats with cystitis/SBU ($P = .008$), UO and healthy cats ($P = .019$), UO and cats with cystitis/SBU ($P = .018$)

FIG 5.



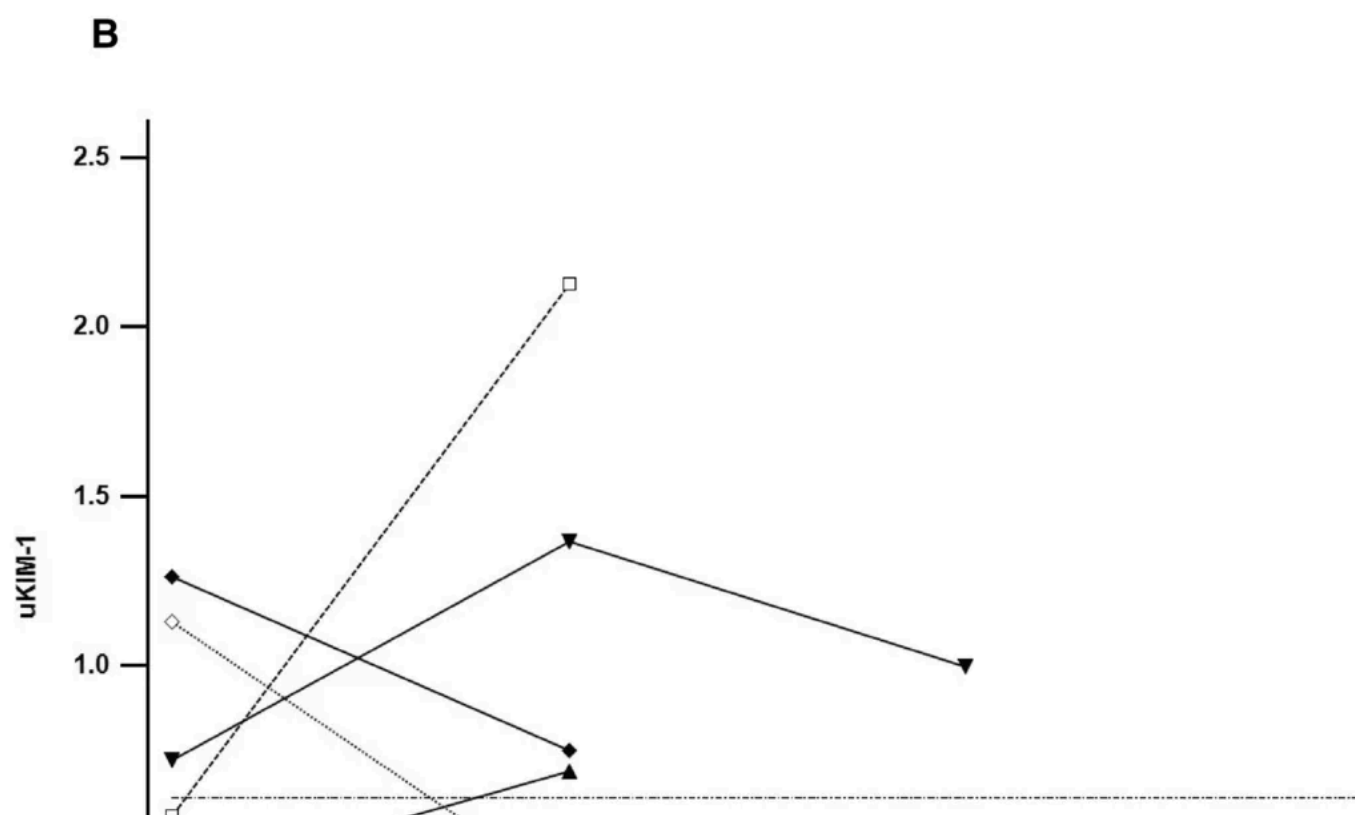
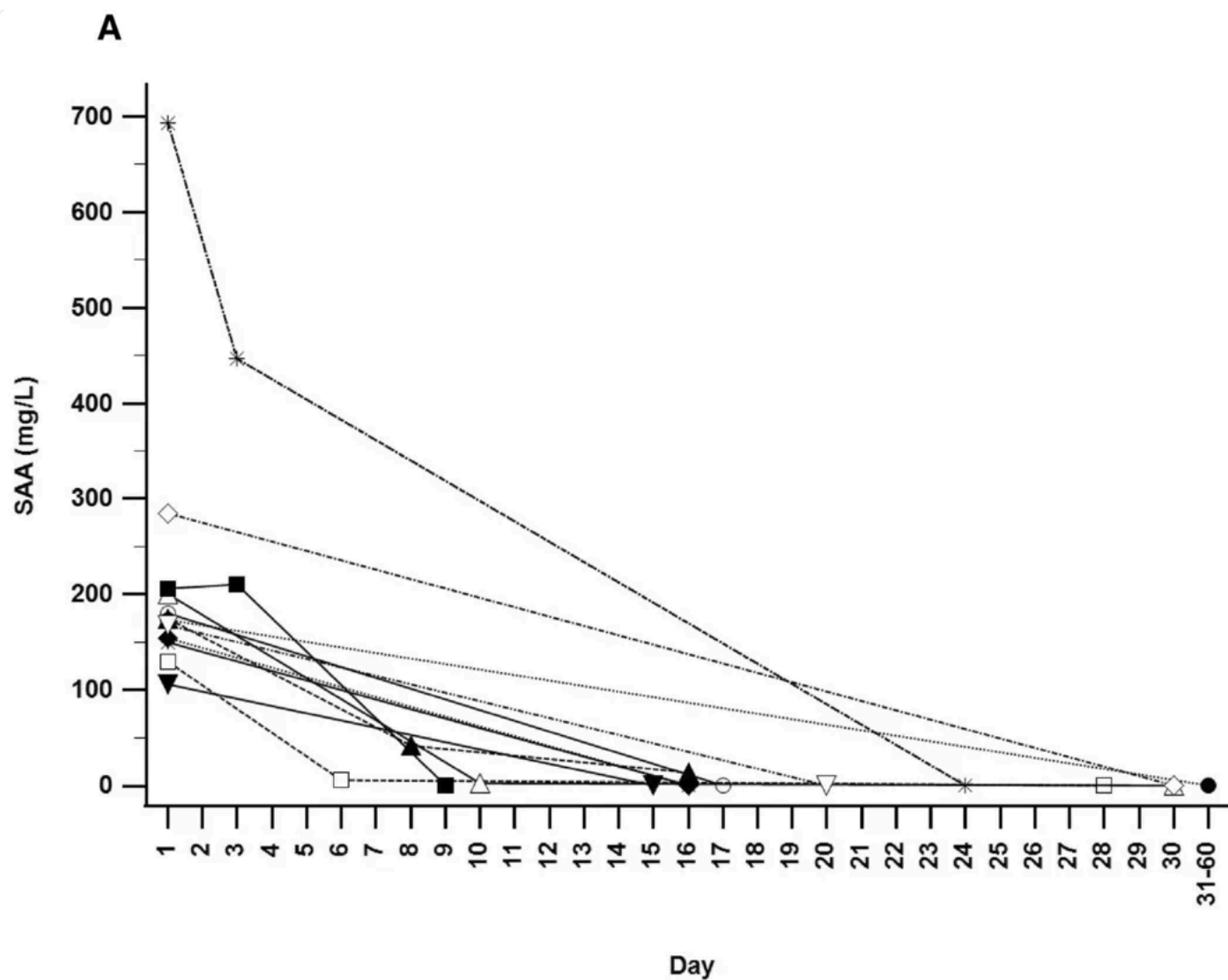
[Open in a new tab](#)

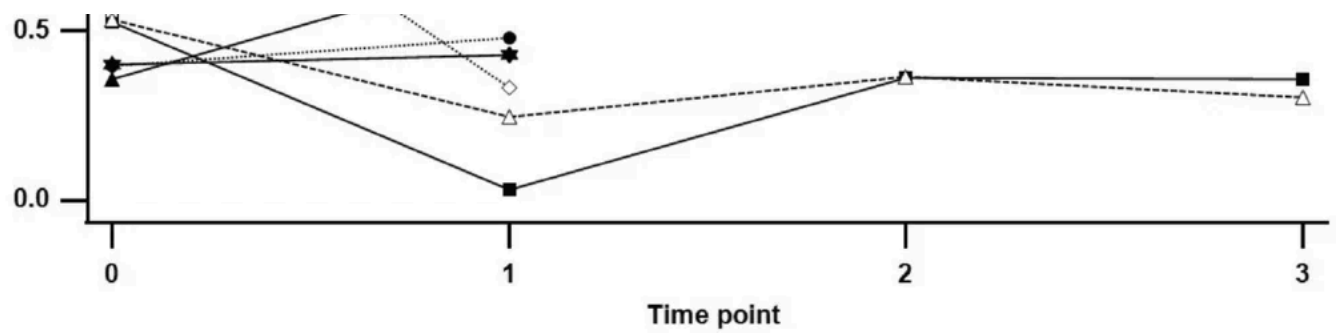
Urinary kidney injury marker-1 (uKIM-1) in 11 clinically healthy cats, 20 cats with confirmed (PN) or presumptive pyelonephritis (pPN), 12 cats with ureteral obstruction (UO) and 11 cats with cystitis or subclinical bacteriuria (SBU). Horizontal lines represent medians and 25th to 75th percentiles. uKIM-1 concentrations are normalised to urine specific gravity of 1035 and presented as control/test ratios. Median uKIM-1 was significantly different between PN/pPN and healthy cats ($P = .009$), PN/pPN and cats with cystitis/SBU ($P = .008$), UO and healthy cats ($P = .019$), UO and cats with cystitis/SBU ($P = .018$).

Follow - up SAA and urinary KIM - 1

Follow-up SAA concentrations were available for 14 cats with PN or pPN after treatment, some of which had a follow-up SAA concentration measured more than once. In 12/14 cats, SAA was completely normalised. In the remaining two cats, SAA was reduced to 5.8 and 13 mg/L at the last follow-up at Day 6 and Day 16, respectively (Fig [6A](#)). Follow-up uKIM-1 values were available for nine cats with PN or pPN (Fig [6B](#)).

FIG 6.





[Open in a new tab](#)

(A) Serum amyloid A (SAA) in consecutive samples from nine cats with confirmed pyelonephritis (solid lines and solid symbols) and five cats with presumptive pyelonephritis (dotted lines and open symbols). Day 1 is the day of inclusion. (B) Urinary kidney injury marker-1 (uKIM-1) in consecutive samples from six cats with confirmed pyelonephritis (solid lines and solid symbols) and three cats with presumptive pyelonephritis (dotted lines and open symbols). The horizontal dotted line represents the upper end of the interval for healthy cats. The time interval between time points was: T0–T1: median 18 days (12 to 137 days), T1–T2: median 22 days (5 to 37 days), T2–T3: 90 to 630 days.

DISCUSSION

The main aim of our study was to assess the ability of SAA concentration and leukocyte ratios to discriminate cats with PN from cats with other urological conditions constituting relevant differential diagnoses. We found a substantial (21- to 169-fold) increase in SAA in all cats with PN, and in all but one cat with pPN. Furthermore, we found minimal overlap in SAA concentrations between cats with PN and cats with other urological conditions. Serum amyloid A proved to be a more sensitive marker of inflammation than the clinical signs, as only 48% of the pyelonephritis cats (PN + pPN) were pyretic (cut-off 39.2°C). During our data processing, another study supporting the use of SAA as a biomarker in cats with PN was published from France (Kurtz et al., [2024](#)). The study population was overall fairly similar to ours, though the group of cats with other urological diseases was wider, and the classification methods varied (classified by panel). When comparing cats with PN/pPN to cats with other urological diseases, Kurtz and coworkers also found a significantly higher SAA concentration in cats with PN, and interestingly, they found an optimal cut-off value of 51.3 mg/L. These results are consistent with ours, indicating reproducibility and robustness of the SAA assay despite different laboratory set-ups, and thus strengthening our confidence in the conclusions. At first glance, our findings and those of Kurtz

et al. appear to conflict with previous studies on SAA in cats with CKD and lower urinary tract disease. However, this discrepancy mainly reflects differences in cut-off values applied. Previous studies (Javard et al., [2017](#); Sasaki et al., [2003](#); Yuki et al., [2020](#)) reported systemic inflammation and increased SAA in cats with CKD and/or lower urinary tract disease, using a cut-off of ≤ 5 mg/L, the upper limit for healthy cats. When applying a higher threshold of 50 mg/L – an approximation of the cut-offs calculated in our study and the study by Kurtz and coworkers – most CKD and lower urinary tract cases fall below this cut-off. In Sasaki et al.'s short communication, an SAA concentration > 50 mg/L was found in only 2/27 cats with lower urinary tract disease and in 12/75 with renal failure (Sasaki et al., [2003](#)). No details were provided on disease duration, concurrent upper urinary tract involvement for the cats with lower urinary tract disease, or comorbidities. Yuki et al. reported significantly higher SAA concentrations in cats with CKD and lower urinary tract disease than in healthy cats, but individual values were not shown. Median SAA concentration in lower urinary tract disease was 0 mg/L (IQR: 0 to 53.8), and in CKD, the IQR was 0 to 5.9 mg/L, suggesting most concentrations fell below 50 mg/L (Yuki et al., [2020](#)). Likewise, Jarvard et al. reported a mean SAA concentration of 8.1 mg/L (SD 6.5) in 38 cats with CKD, supporting the notion that inflammation in chronic renal disease does not elicit a high magnitude acute phase response (Javard et al., [2017](#)). Degenhardt et al. studied SAA in cats with renal azotaemia (acute kidney injury (AKI), CKD and acute-on-chronic kidney disease (AOC)). Of 44 cats, 15 had SAA concentrations > 50 mg/L; four had PN and the remaining 11 cats were diagnosed with AOC, AKI or CKD plus a variety of comorbid conditions, some of which are well known to be associated with systemic inflammation (pyometra, perforated intestinal foreign body, diabetic ketoacidosis) (Degenhardt et al., [2023](#)). In cats with CKD, an SAA increase was associated with comorbid conditions in all cases, whereas the cause of a high SAA concentration was less obvious in cats with AKI or AOC.

To conclude this argument, causes of increased SAA concentration in cats with azotaemia are obviously not restricted to PN, and this non-specific acute phase protein should be interpreted in the context of other clinical and paraclinical findings. However, in cats with azotaemia and bacteriuria, an increased SAA concentration > 50 mg/dL can be considered indicative of PN, unless another apparent cause of inflammation is identified. Likewise, PN can be considered unlikely in cats with azotaemia and bacteriuria if SAA lies below the 50 mg/L threshold. This is particularly helpful in a clinical setting, as SBU or cystitis in cats with underlying CKD is considered difficult to discriminate from PN due to the unspecific clinical presentation of the latter.

This study further identified that the SAA concentration in cats with PN decreases with antimicrobial treatment. The clinical implications include a potential role for SAA to guide the duration of treatment and thus target antimicrobial therapy to the individual cat. However,

tailoring therapy based on SAA response was not examined in our study, and the SAA concentration as a response biomarker during treatment was not systematically assessed.

Neutrophil:lymphocyte ratio was considered a potentially interesting biomarker as it combined parts of both the innate and adaptive immune system. The NLR was found to be significantly higher in cats with PN compared to other cats. However, values overlapped between groups, and a positive test result only modestly increased the likelihood of PN. It is doubtful that NLR provides any added discriminatory value when compared to other components of the blood count, such as neutrophilia and an increased number of band neutrophils. Neutrophil:lymphocyte ratio may potentially play a part in monitoring of treatment in cats with increased ratios; this, however, was not tested in our study. Of interest, NLR was also assessed in the study by Kurtz and coworkers and was not found to be significantly different between groups (Kurtz et al., [2024](#)).

The second aim of our study was to assess tubular damage in cats with PN, and we detected mild to marked uKIM-1 elevations in more than 50% of cats with PN and UO, confirming that proximal tubular injury is common in these populations of cats. We also detected large variations in uKIM-1 concentrations in cats with PN and UO, which can be attributed to several factors. Production of KIM-1 by proximal renal tubular cells and its shedding into urine depend on the severity, extent and duration of injury. Severe AKI results in markedly increased KIM-1 in the distal segment of the proximal tubular cells after 24 hours and occurs before decreases in USG (Bland et al., [2017](#)). But severe injury can also cause cell swelling, cell death, and obstruction of urine flow. If this occurs unilaterally, low uKIM-1 from a non-affected kidney may mix with smaller volumes of high uKIM-1 from an injured kidney and result in overall mildly to markedly increased, or even normal, uKIM-1 values, which may change as urine flow is re-instituted and/or the epithelium regenerates. In our study, four out of nine cats, from which consecutive samples were available, displayed marked changes in uKIM-1 between initial presentation and the first follow-up visit (Fig [6](#)), stressing that sequential uKIM-1 measurements are likely more useful than single samples to measure the extent of injury and repair.

Another factor of importance in our study concerns the anatomic localisation of the inflammation detected. Ascending infection may be confined to the renal pelvis (pyelitis) or extend into the renal parenchyma (pyelonephritis), conditions that cannot be reliably differentiated in clinical practice. Anatomically, KIM-1 is expressed predominantly by proximal tubular epithelial cells (Bland et al., [2017](#)). Therefore, infections confined to the renal pelvis may not provoke significant KIM-1 expression. The importance of anatomic localisation was highlighted in a preclinical rodent model of AKI, in which uKIM-1 was increased in rats with PN-associated AKI, but not in rats with acute

PN alone (Skowron et al., [2018](#)). Based on histopathology, the authors of that study concluded that localisation of renal inflammation differed between rats with and without uKIM-1 increases.

Lastly, disease chronicity may involve fibrotic or atrophic tubules, leading to reduced or absent KIM-1 production (Zhang et al., [2008](#)). This may be of relevance in our population of cats with PN or UO suffering from multiple coexisting conditions of unknown duration, including CKD. Declining capacity of KIM-1 production with advanced CKD was also proposed in a recent study of cats, in which uKIM-1 concentrations of cats with CKD did not differ from those of healthy cats (Kornya et al., [2025](#)).

There are some limitations to the present study. The sample size was not derived from a sample size calculation but was based on feasibility (PN cases confirmed by pyelocentesis are rare) and available funding. We used a case–control design for enrolment of cats rather than consecutive enrolment of all cats presenting with urological disease. Though case–control enrolment is commonly used in diagnostic studies, it carries an inherent risk of selection bias. However, in our study, the control group of cats with urological disease resembled cats with PN or pPN in terms of complexity (azotaemia and presence of comorbidities), thus increasing the generalizability of the results.

Two different SAA assays were used in the present study, the LZ-SAA and the VET-SAA, the latter used also in the study by Kurtz and coworkers (Kurtz et al., [2024](#)). Published comparisons of the LZ-SAA and VET-SAA assays report high concordance in classifying samples as normal or increased, despite modest differences in absolute values (Waugh et al., [2022](#)). The transition to the VET-SAA assay in June 2022 resulted in SAA quantification for ten animals, six of which had very high SAA values (199 to 848 mg/L) and the remaining 4 had SAA values <5 mg/L. Therefore, while the assay transition may have introduced minor analytical variation, it is unlikely to have impacted the interpretation of results in this study.

Serum samples from three cats were stored for an extensive time period before analysing SAA. In cats, SAA has been shown to remain stable for prolonged periods at –20°C (Ceron et al., [2005](#)). In the present study, samples were stored continuously at –80°C with minimal freeze–thaw cycles. Under these conditions, significant degradation of feline SAA is considered unlikely. Finally, the SAA concentrations in the three samples stored for 9 years were 199, 640 and 848 mg/L, respectively. Even if hypothetically some SAA protein degradation took place during storage it would not have altered the conclusions of this study.

In conclusion, we found that SAA increases in cats with PN and decreases in response to therapy. When applying a threshold of 50 mg/L, SAA increases the likelihood of pyelonephritis substantially in cats with urological diseases. In addition, our results indicate that proximal tubular injury is common in cats with PN and cats with UO. The course of tubular injury is best assessed by consecutive testing; however, the clinical implications of uKIM-1 alterations remain to be investigated.

Author contributions

L. R. Jessen: Conceptualization; investigation; funding acquisition; writing – original draft; methodology; validation; visualization; writing – review and editing; project administration; data curation; supervision; software; formal analysis. **R. Langhorn:** Investigation; writing – review and editing; writing – original draft. **N. Dupont:** Investigation; writing – review and editing. **A. S. Gravgaard:** Investigation; writing – review and editing. **N. Bak-Jacobsen:** Investigation; writing – review and editing. **B. J. Reezigt:** Investigation; writing – review and editing. **L. N. Nielsen:** Writing – original draft; methodology; validation; formal analysis; writing – review and editing; software. **D. Bienzle:** Investigation; writing – original draft; writing – review and editing; supervision; formal analysis; software; validation; methodology. **F. S. Moberg:** Investigation; writing – review and editing. **A. L. Jensen:** Writing – original draft; methodology; formal analysis; data curation; writing – review and editing; software.

Conflict of interest

D. Bienzle is a co-inventor of the feline KIM-1 assay. None of the authors of this article has a financial or personal relationship with other people or organisations that could inappropriately influence or bias the content of the paper.

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Supporting information

Table S1. STARD 2015 reporting guidelines.

[JSAP-67-810-s004.docx](#) (22.1KB, docx)

Table S2. Comorbidities in 14 cats with confirmed PN, 8 cats with presumptive PN, 35 cats with other urological diseases (10 with CKD, 14 with UO, 11 with SBU/cystitis) and 14 healthy cats.

[JSAP-67-810-s003.docx](#) (16.9KB, docx)

Table S3. Key features of 35 cats with other urological diseases.

[JSAP-67-810-s001.docx](#) (18.1KB, docx)

Table S4. Serum amyloid A, neutrophil:lymphocyte ratio and monocyte:lymphocyte ratio measurements in cats with chronic kidney disease, cats with ureteral obstructions and cats with bacterial cystitis or subclinical bacteriuria.

[JSAP-67-810-s002.docx](#) (16.2KB, docx)

The last two authors have shared last authorship.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

Table S1. STARD 2015 reporting guidelines.

[JSAP-67-810-s004.docx](#) ^(22.1KB, docx)

Table S2. Comorbidities in 14 cats with confirmed PN, 8 cats with presumptive PN, 35 cats with other urological diseases (10 with CKD, 14 with UO, 11 with SBU/cystitis) and 14 healthy cats.

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